Backend Engineer Roadmap (2025 - 2028)

2025 – Core Foundations

- Learn DBMS + SQL deeply (queries, indexing, joins, transactions).
- Study Operating Systems basics: processes, threads, memory management.
- Learn Computer Networks: HTTP, DNS, TCP/IP, load balancing.
- Strengthen DSA (Data Structures & Algorithms).

2025–26 – Backend Frameworks

- Master one backend framework: Node.js (Express/NestJS) OR Django/FastAPI.
- Learn authentication & authorization (JWT, OAuth).
- Build REST APIs and GraphQL APIs.
- Practice middleware, ORM, and error handling.

2026 – Databases & DevOps Foundations

- SQL: PostgreSQL/MySQL (transactions, query optimization).
- NoSQL: MongoDB, Redis basics.
- Learn Docker for containerization.
- Start with Kubernetes for orchestration.
- Linux basics: shell scripting, monitoring.

2026–27 – Cloud & Scalability

- Learn AWS (EC2, S3, RDS, Lambda, API Gateway, Load Balancers).
- Infrastructure as Code: Terraform basics.
- Deploy apps on cloud (AWS/GCP).
- Implement CI/CD pipelines (GitHub Actions, Jenkins).

2027 – Microservices & System Design

- Understand microservices architecture (service discovery, API gateway).
- Learn messaging queues (Kafka, RabbitMQ).
- REST vs gRPC communication.
- System Design practice: design Twitter, YouTube, E-commerce systems.
- Learn caching, sharding, load balancing, rate limiting.

2027–28 – Advanced Topics

- Distributed Systems concepts: CAP theorem, eventual consistency.
- Observability: logging, monitoring, tracing.
- Security: OWASP, input validation, encryption.
- Scalable project: build & deploy real-world apps with microservices.

Projects to Build

- Blogging system (monolith → microservices).
- E-commerce backend (users, inventory, orders, payments).
- Real-time chat app (WebSockets + Redis).
- Deploy apps using Docker + Kubernetes + AWS.

Extra Skills

- Search Engines (Elasticsearch).
- Message Brokers (Kafka, RabbitMQ).
- Git mastery + CI/CD pipelines.
- Communication & soft skills for interviews.

Recommended Resources

DBMS & SQL

- Book: Database System Concepts – Silberschatz, Korth, Sudarshan
- Course: SQL for Data Science (Coursera)
- Docs: PostgreSQL official documentation

Operating Systems & Networks

- Book: Operating System Concepts – Silberschatz
- Book: Computer Networking: A Top-Down Approach – Kurose, Ross
- Course: CS50's Computer Science (Harvard)

Backend Frameworks

- Docs: Express.js official docs
- Docs: Django & Django REST Framework
- Course: Node.js, Express & MongoDB (Udemy)

DevOps (Docker, Kubernetes, Linux)

- Book: Docker Deep Dive – Nigel Poulton
- Course: Kubernetes for Beginners (KodeKloud)
- Docs: Linux Command Line (Linux.org)

Cloud & CI/CD

- Course: AWS Certified Solutions Architect – Associate
- Docs: AWS official documentation
- Tutorial: GitHub Actions official docs

Microservices & System Design

- Book: Designing Data-Intensive Applications – Martin Kleppmann
- Course: System Design Primer (GitHub)
- Course: Microservices with Node.js and React (Udemy)

Advanced Topics

- Book: Site Reliability Engineering – Google SRE team
- Course: Distributed Systems (MIT OCW)
- Docs: OWASP Security Guidelines